

Maciej Tkacz

Simulation Engineer - Robotics & Physics-Based Simulation

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PROFESSIONAL PROFILE

Simulation engineer profile built on commercial physics-based simulation of mechanical and robotic systems - and on building the physical machines those simulations describe. Currently a Robotics Engineer at ASTOR, building digital twins of production lines and robot simulations, programming Kawasaki and Epson robots, and designing and commissioning robot demonstration stations, so I see both the model and the machine it is supposed to match. Previously an Application Engineer at AIAutomation for a year and a half: digital twins of production workcells for automotive clients, validating and tuning kinematics, joint limits, reach and motion constraints, integrating customer CAD models into the simulation environment, and writing Python scripts that automate simulation logic and improve simulation performance. Mechanical Engineering student at Cracow University of Technology (dynamic systems modelling, control, analytical mechanics) and certified Software Technician, so the physics side and the scripting side come from the same background. I also designed and built a SCARA robotic arm end-to-end, which gives me a machine of my own to validate simulated behaviour against. My simulation stack so far is Visual Components rather than Omniverse. My next project is rebuilding that SCARA arm inside a USD-based simulator to compare simulated and real behaviour. Based in Krakow, available full-time.

KEY SKILLS

- Python for simulation: scripting simulation logic and behaviour, process-optimization scripts, automation of repetitive simulation tasks, calculation and data-processing tools; C++ and Arduino/C for hardware-level control.
- Physics-based simulation of mechanical systems: validating and tuning kinematics, joint limits, reach, collisions and motion constraints; university coursework in dynamic systems modelling, control, mechatronics and analytical mechanics.
- CAD models inside simulation environments: preparing, cleaning and optimizing 3D geometry from SolidWorks, Autodesk Inventor, Fusion 360 and AutoCAD for use in simulation, balancing visual fidelity against simulation performance.
- Robotics simulation & digital twins: digital twins of production lines and workcells - robot paths, process logic, collision checking, cycle flows, layout optimization and virtual validation before physical implementation; Visual Components used commercially for automotive clients.
- Sim-to-real grounding: I program, build and commission the physical robot cells I also model, and I designed and built a SCARA arm I control in Python/C++ - a direct reference for simulated kinematics, gear ratios, joint limits and actuator behaviour.
- Simulation stack roadmap: moving into NVIDIA Isaac Sim / Omniverse and the USD asset pipeline, with my own SCARA arm as the planned porting target; ROS from personal robotics projects; MATLAB from university.
- Engineering practice: version control with Git, technical documentation of simulation workflows, assumptions and constraints, and work with customer documentation and engineering standards.
- Industrial robots: Kawasaki and Epson programming - certified courses, teach pendant workflows, application cycles and robot cell testing.

SELECTED SIMULATION & ROBOTICS PROJECT

SCARA Robotic Arm - Physical Prototype and Kinematic Model | Personal R&D Project

- Designed and built an end-to-end SCARA robotic arm: mechanical concept, CAD modelling, kinematic layout, gear ratios, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Defined and verified the physical parameters that drive simulated behaviour - link geometry, joint limits, gear ratios and actuator torque - against the assembled machine.
- Developed Python/C++ control software implementing forward and inverse kinematics, giving a real measurement point for comparing modelled and physical motion.
- Optimized components for 3D printing, including PET-G and carbon-fibre reinforced materials, with focus on stiffness, assembly constraints and rapid iteration.
- Planned next step: porting this arm into a USD-based simulator to run identical motion sequences in simulation and on the physical machine and compare the results.

PROFESSIONAL EXPERIENCE

Robotics Engineer | ASTOR, Krakow | Jul 2026 - present

- Build digital twins of production lines and robot simulations, validating reach, motion sequences and process behaviour before physical implementation.
- Program Kawasaki and Epson industrial robots, so the motion I model in simulation is motion I also implement and verify on the physical machine.
- Design, build and commission robot demonstration stations presenting Kawasaki and Epson robots in different applications - mechanical assembly, hardware-software integration, programming and bring-up.
- Work across mechanics, electrics and robot software when testing and troubleshooting robot cells, which is where simulated and real behaviour get compared in practice.

Application Engineer | AIAutomation | Jan 2025 - May 2026

- Built and maintained physics-based simulations and digital twins of production workcells in Visual Components for automotive clients.
- Validated and tuned robot kinematics, reach, joint limits, motion constraints and collision behaviour so that simulated cells matched real production requirements.
- Integrated customer CAD models from SolidWorks and Autodesk Inventor into the simulation environment, preparing and optimizing 3D geometry for simulation use and performance.

- Developed Python scripts to automate simulation logic, optimize processes and improve simulation behaviour and performance.
- Implemented robot logic, motion sequences, robot paths, cycle flows and virtual process validation, and optimized workcell layouts.
- Documented simulation assumptions, constraints and workflows, and analyzed customer documentation and engineering standards to keep simulated behaviour aligned with the physical process.

3D Printing and CAD Design Specialist | Cubic Inch Additive Manufacturing, Piaseczno | Jun 2023 - Aug 2023

- Operated and serviced FDM, MJF and SLA 3D printers, supervising process parameters, post-processing and quality control.
- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing and rapid prototyping.
- Supported implementation testing for a new SLA technology, documenting progress and technical observations.
- Coordinated production tasks in a 10-person project team, balancing quality, manufacturability and delivery constraints.

Robotics and 3D Printing Intern | ASTOR Robotics Center, Krakow | May 2022

- Assembled mechanical equipment and 3D-printed components for Kawasaki robots and Astorino educational robot platforms.
- Programmed Kawasaki industrial robots using teach pendant workflows and created basic robot motion programs.
- Tested robot and workstation operation, gaining practical exposure to robot setup, safety and hardware integration.

Web Application Development Intern | Souczek Design Studio Reklamy i Druku, Kielce | Jul 2021

- Solved technical problems in JavaScript, including implementation of an interactive order form.
- Tested and deployed web functionality in a team environment according to customer guidelines.

EDUCATION

- Cracow University of Technology - Mechanical Engineering, Engineer's degree in progress (Oct 2024 - present). Relevant coursework: dynamic systems modelling, automation/control, analytical mechanics and mechatronics.
- PKMechPower Student Research Group, Mechanical Section - CAD and 3D printing tasks, including driver seat and brake-system components; mass and strength-oriented design optimization.
- Zespol Szkol Informatycznych im. gen. Jozefa Hauke-Bosaka, Kielce - Software Technician (Sep 2019 - Apr 2024).

CERTIFICATES

- Kawasaki robot operation and programming - integrator course with certificate, ASTOR Robotics Center (May 2022).
- Epson industrial robot programming - certified course.
- Python programming courses and self-directed Python/C++ development in robotics and automation projects.

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